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McLennan, Self-Confirming Equilibrium, Incomplete Information, and Learning from Chosen Actions

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Background

Self-confirming equilibrium grew out of the idea that players may learn only from what they actually experience, not from all payoff-relevant facts in the game. In games with incomplete information, this matters even more because players must form beliefs both about others' actions and about hidden states or types, and some wrong beliefs can survive if play never reveals them. (LLM Memory)

The basic background is the contrast with Nash equilibrium. In Nash equilibrium, each player's strategy is optimal given correct beliefs about the parts of the game that matter for that strategy, and those beliefs are mutually consistent. Self-confirming equilibrium weakens this by allowing players to hold wrong beliefs about things they never observe, as long as the beliefs are correct on the path of play and each chosen action is optimal given those beliefs. This idea is tied to adaptive play and learning: if feedback is limited, players can learn to act in stable ways without ever learning the whole game. In incomplete-information settings, the same logic applies but beliefs now concern not only opponents' behavior but also hidden payoff states, signals, or types. That is why work in this area often focuses on what the player can observe after choosing an action, what data the feedback generates, and whether different hidden models of the world look the same from the player's own experience. McLennan's contribution is commonly placed in this line of work on equilibrium under limited feedback and imperfect knowledge, linking stable play to what agents can infer from chosen actions rather than to full correct specification of the environment. (Model-Generated)

Core definition of self-confirming and conjectural equilibrium under incomplete information

In self-confirming equilibrium, each player best responds to their own beliefs, and those beliefs only need to fit what that player actually observes from the play that occurs. In incomplete-information games, this means beliefs can be wrong about hidden states, types, off-path actions, or unused contingencies, as long as the player's realized evidence does not refute them. (6 sources)

A useful core definition is this: a profile is a self-confirming equilibrium if each player's strategy is a subjective best reply to that player's beliefs, and those beliefs are confirmed by the observations generated along the realized path of play. This is the classic weakening of Nash equilibrium: beliefs need to be right where experience gives feedback, but they need not be right at off-path information sets or about events the player never sees. That is the basic idea in the original self-confirming equilibrium formulation and in later restatements. (Battigalli et al., 2022) (Fudenberg et al., 1993) (Macault et al., 2020) (Dekel et al., 2004)

Under incomplete information, the object of belief is broader. A player may have beliefs not just about opponents' actions, but also about Nature's moves, payoff states, or types, and the equilibrium requirement is

that the chosen strategy be optimal given those beliefs and that the beliefs match the distribution of outcomes or signals that the player observes. This observation-based consistency is central in incomplete-information versions because players may not observe others' actions directly, and may sometimes not even observe their own payoff. (Dekel et al., 2004)

This is also why conjectural equilibrium is often treated as a close relative or refinement of self-confirming equilibrium. The literature uses these labels for profiles with confirmed beliefs and subjective best reply, with conjectural equilibrium especially linked to extensive-form games of incomplete information and settings without a common prior. In that form, the idea is that players act rationally given the information they have about the game's unknown parameters, while their conjectures need only be validated by the evidence their own play produces. (Battigalli et al., 2022) (Macault et al., 2020)

An important implication is that self-confirming equilibrium can be much larger than Nash equilibrium when feedback is coarse. If players observe only limited outcomes, then many different beliefs can fit the same observed data, so stable behavior need not pin down the true model of opponents' play or the true distribution of types. In such cases, players can disagree in ways Nash would not allow, including having different forecasts about an opponent's action distribution because they hold different beliefs about that opponent's type. (Dekel et al., 2004)

This logic extends even to one-player decision problems with uncertainty: equilibrium beliefs are not free, but are disciplined by choice-dependent evidence. So the defining feature is not strategic interaction alone; it is the link between optimal choice and the feedback generated by the chosen action. (Battigalli et al., 2022)

Some formulations also stress that players may hold subjective beliefs with rich dependence patterns, such as beliefs that opponents' strategies are correlated. That does not change the core definition: what matters is still best response under subjective beliefs plus confirmation by observed outcomes. (Sota et al., 2013)

Finally, in recent work on incomplete information with private values, the self-confirming equilibrium set at a given state can be independent of players' interactive knowledge about the full profile of payoff functions. This reinforces the main point: the concept is built around what each player can justify from their own evidence, not around full shared correct understanding of the whole environment. (Battigalli et al., 2024)

Evidence

(Battigalli et al., 2022)

A framework for the analysis of self-confirming policies

> *"In the game-theoretic literature, a strategy profile that satisfies the properties of confirmed beliefs and subjective best reply has been called "conjectural equilibrium" (Battigalli, 1987;Battigalli & Guaitoli, 1988), "self-confirming equilibrium" (Fudenberg et al., 1993) and "subjective equilibrium" (Kalai et al., 1993)(Kalai et al., 1993)"*
>
> *".We extend the analysis of self-confirming equilibria to settings with inherent randomness and possibly infinite spaces of strategies and states of nature"*
>
> *".Second, it clarifies that selfconfirming equilibrium is a genuine equilibrium concept also in a one-person setting, because equilibrium beliefs are disciplined by choice-dependent evidence."*

(Fudenberg et al., 1993)

Self-confirming equilibrium

> *"In a self-confining equilibrium, each player's strategy is a best response to his beliefs about the play of his opponents and each player's beliefs are correct along the equilibrium path of play. Thus, if a self-confirming equilibrium occurs repeatedly, no player ever observes play that contradicts his beliefs, even though beliefs about play at off-path information sets need not be correct. The authors characterize the ways in which self-confirming equilibria and Nash equilibria can differ and provide conditions under which self-confirming equilibria correspond to standard solution concepts. Copyright 1993 by The Econometric Society."*

(Macault et al., 2020)

Social learning in nonatomic routing games

"(Fudenberg et al., 1993) introduced the concept of self-confirming equilibrium, where players' beliefs about other players' actions are required to be correct only on the equilibrium path. As agents play best responses to their beliefs, these may differ substantially from the truth, as long as nothing contradicts the beliefs. Hence, self-confirming equilibria and Nash equilibria of a game need not be the same. Given the sequential nature of the repeated game, self-confirming equilibria may allow players to hold beliefs on other agents that may be inconsistent with rational behavior. Battigalli and Guaitoli (1997) refined self-confirming equilibrium by proposing the concept of conjectural equilibrium for extensive form games of incomplete information without a common prior. At a conjectural equilibrium, players are assumed to behave rationally given the information they have about the game parameters"

".Namely, we study the steady states of social learning dynamics where players myopically bestrespond to the information obtained by previous generations. We show that this information is correct for the edges of the network that have been explored along the equilibrium path, but may remain incorrect for other edges. Thus, some paths that would be used under full information may remain unused forever."

(Dekel et al., 2004)

Learning to play Bayesian games

"This equilibrium concept requires that each player's strategy is optimal given the player's beliefs about opponents' strategies and Nature's moves, and that these beliefs are consistent with the distribution of outcomes in the game that the player observes. In Section 2 we give a formal definition that extends the notion to allow for different possible maps from outcomes of the game to observations, in particular allowing for the case where players do not observe the actions taken by others or their own payoffs"

".When players observe less than the outcome, more beliefs are consistent with the players' observations, so a Nash equilibrium without a common prior may be a self-confirming equilibrium, but only because the set of self-confirming equilibria is then large (and therefore may well include many outcomes that are not Nash equilibria)"

".We will not formally model the dynamics of learning, but will appeal informally to the idea that a steady state of a belief-based learning process must be a self-confirming equilibrium (Fudenberg et al., 1993)). Thus, our focus is on how the information that players observe at the end of each round of play determines the set of self-confirming equilibria, and how these equilibria relate to the Nash equilibria of the game"

".One way of summarizing this example is to say that although Nash equilibrium requires two players to agree about an opponent's strategy, these players can have different forecasts about the distribution of that opponent's actions if they have different beliefs about the distribution of that player's type."

(Sota et al., 2013)

How Albanian Private Universities can use Game Theory for Optimization of Scholarship Offers

"Fundenberg and Lebine (1993) introduced the notion of "self-confirming equilibrium" for finite subjective games. Each player in such a game chooses a subjective strategy to maximize his/her expected utility function, given his/her subjective beliefs about opponents' strategies. These beliefs allow the possibility that the opponents' strategies are positively (or negatively) correlated."

(Battigalli et al., 2024)

Sophisticated reasoning, learning, and equilibrium in repeated games with imperfect feedback

"Indeed, in an environment with possibly incomplete information and private values, the SCE set at any given state of nature is independent of players' interactive knowledge of the profile of payoff functions. 2"

Observational equivalence and feedback structure

What matters for self-confirming equilibrium is not whether beliefs are globally true, but whether the feedback a player gets could also have been generated by those beliefs. When different hidden states, type distributions, or opponent strategies produce the same observed evidence, they are observationally equivalent, and that lets incorrect beliefs survive. (6 sources)

A clean way to see the role of feedback is to ask what data a player actually gets from play. If feedback consists only of observed actions, realized terminal nodes, local signals, or own payoff, then beliefs are tested only against that limited evidence, not against the full game. This is why self-confirming equilibrium is naturally tied to a feedback structure: equilibrium beliefs must fit the observations induced on the realized path, but need not fit unobserved contingencies. Dekel, Fudenberg, and Levine make this point in a stark way: when players only observe realized play, they can learn the equilibrium path while never getting evidence against off-path mistakes (Dekel et al., 1999). Fudenberg, He, and Imhof later restate the same basic lesson more generally: non-Nash behavior can persist because the data agents collect may never challenge false beliefs (Fudenberg et al., 2019).

This leads directly to observational equivalence. Two models of opponents' behavior, payoff states, or type distributions are observationally equivalent for a player if they imply the same distribution of whatever that player can observe after choosing an action. Once equivalence is defined in that player-specific way, self-confirmation only rules out beliefs that are contradicted by the induced feedback; it does not rule out wrong hidden stories that generate the same evidence. Schipper's formulation makes this especially explicit: along the realized play, the player is certain only of behavioral strategies of opponents and Nature that are observationally equivalent to what was observed, and beliefs at reached information sets remain consistent with what has been and will be seen on that path (Schipper, 2017). Battigalli and coauthors make the same point in broader environments with limited awareness and local information: agents best respond to conjectures that are consistent with the feedback they receive, and that feedback need not be fully revealing (Battigalli et al., 2018).

In incomplete-information settings, the same logic becomes sharper because hidden type distributions or states can often be observationally equivalent under the mechanism or action profile that is actually played. Then many distinct conjectures can support the same behavior. Recent mechanism-design language states this plainly: a mechanism is self-confirming if it is sustained by at least one belief compatible with observed behavior, and the concept can be permissive precisely because many type distributions are observationally equivalent under that mechanism (Feng et al., 2026). This is the sense in which McLennan-style learning from chosen actions departs from full-information equilibrium reasoning: the chosen action filters the evidence, and that filter can make very different underlying worlds look the same to the decision maker (Model-Generated).

The feedback structure also helps explain why refinements matter. Standard self-confirming equilibrium often uses only consistency with the equilibrium distribution over terminal outcomes, which is weak when terminal feedback is coarse. Fudenberg, Kreps, and Levine note that this can ignore information players already have about opponents' payoffs, and they motivate rationalizable self-confirming equilibrium as a way to combine feedback consistency with stronger inference from known structure (Fudenberg et al., 2015) (Dekel et al., 1999). So, for the query's themes, the main connection is: incomplete information creates hidden variables, chosen actions determine what evidence is seen, and observational equivalence tells us when learning from that evidence can fail to identify the true game.

Evidence

(Dekel et al., 1999)

Payoff Information and Self-Confirming Equilibrium

> *"where players have no a priori information about opponents' play or payoffs, and each time the game is played they observe only the actions played by their opponents.*
>
> *Intuitively, self-confirming equilibrium requires only that players correctly forecast the actions opponents will take along the equilibrium path, but does not require that their offpath beliefs are correct. When the only information players have is the observed play in the game, they will never receive evidence that their forecasts of off-path play are incorrect."*

(Fudenberg et al., 2019)

Learning Theory and Heterogeneous Play in a Signaling-Game Experiment

> *"A key reason why non-Nash behavior can arise in equilibrium is the possibility that incorrect beliefs are not challenged by the data that agents collect over time."*

(Schipper, 2017)

Self-confirming Games: Unawareness, Discovery, and Equilibrium

> *"Condition (ii) consists of two properties. First, at the end of the game the player is certain of observationally equivalent behavioral strategies of opponents and nature that allow her to reach the particular end of the game. That is, terminal beliefs are consistent with what has been observed during play (and hence at the end of the play). Second, beliefs do not change during the play. That is, beliefs at any information set reached during the play are consistent with what is observed at any point during the play and in particular with what is observed at the end of the game. Again, the idea is that everything that could have been learned on this path*

has been learned already in the past."

(Battigalli et al., 2018)

Learning and Selfconfirming Equilibria in Network Games

"We analyze a notion of equilibrium which is broader than Nash equilibrium. Recall that our approach allows for the possibility of agents who are unaware of the full game around them. In equilibrium, agents best respond to conjectures consistent with the feedback that they receive, which is not necessarily fully revealing. We believe that this approach fits well to a networked environment where agents' knowledge and the information they receive are only local.6"

(Feng et al., 2026)

Self-Confirming Mechanisms

"This corresponds to the notion of selfconfirming equilibrium: the mechanism is sustained by at least one belief compatible with observed behavior"

".A fourth, and more substantive, issue is that the equilibrium concept can be per-missive, since a wide range of mechanisms may be self-confirming given the many type distributions that are observationally equivalent under these mechanisms."

(Fudenberg et al., 2015)

Rationalizable Partition-Confirmed Equilibrium

"Self-confirming equilibrium (SCE) formalizes the idea that incorrect off-path beliefs can persist for settings where players observe the terminal node of the game each time it is played, and the only restrictions placed on the players' beliefs is that they be consistent with the equilibrium distribution on terminal nodes. However, because SCE places no a priori restrictions on the players' beliefs, it does not capture the idea that players use prior information about opponents' payoff functions to predict the opponents' play. To capture such predictions, (Dekel et al., 1999) (hereafter DFL) define "rationalizable self-confirming equilibrium" (RSCE), which requires that players make certain inferences based on their knowledge of the other players' payoff functions and observation structure."

Learning from chosen actions and incomplete learning dynamics

In this line of work, what agents learn depends on the actions they actually choose, because those choices determine which payoffs or outcomes they get to observe. This can make learning stop at a self-confirming point: behavior is stable and beliefs fit observed data, but wrong views about untried actions or unseen outcomes remain. (3 sources)

A key link between self-confirming equilibrium and learning is endogenous information acquisition: players do not observe all payoff-relevant facts at once, but only the consequences of the actions they actually take. Because of that, learning can converge to a stable fixed point in which current actions are optimal given current beliefs, while beliefs about unchosen actions stay wrong simply because no data are ever generated to test them. Wu et al. make this point directly, noting that their learning fixed points can differ from complete-information equilibrium because estimates of unobserved game outcomes may be incorrect, and these mistakes may never be corrected since information is acquired through the strategies chosen in play. They explicitly connect this to the classic self-confirming equilibrium idea in extensive-form games. (Wu et al., 2021) (Fudenberg et al., 1993)

This gives a simple learning foundation for self-confirmation. If players repeatedly choose the same actions, they keep seeing only the outcomes attached to those actions. Then their beliefs about the realized path can become accurate, even while their beliefs about foregone actions, off-path continuations, or hidden payoff distributions remain false. In that sense, incomplete learning is not an accident but a built-in consequence of choice-dependent feedback. Wu et al. state this sharply: complete learning need not occur because beliefs can assign wrong payoffs to strategies not taken, and those wrong estimates persist precisely because the evidence used for updating is generated endogenously by chosen strategies. They place this phenomenon in the same family as self-confirming equilibrium learning. (Wu et al., 2022) (Fudenberg et al., 1993)

For the incomplete-information query, the point is even stronger. When the player must learn not only about rivals' play but also about hidden states, types, or payoff distributions, chosen actions act like a filter on inference. Some actions reveal a lot; others reveal almost nothing. If play settles on actions with weak feedback, agents may never distinguish between the true environment and an observationally equivalent but

false one. The result is stable behavior with only partial learning: enough learning to support best response on path, but not enough to recover the full game. This is exactly the sense in which McLennan-style reasoning ties equilibrium to learning from chosen actions rather than to full correct beliefs about all contingencies. (Wu et al., 2021) (Wu et al., 2022) (Fudenberg et al., 1993)

So the main lesson is dynamic. Self-confirming equilibrium is not just a static relaxation of Nash; it can arise as the limit of learning processes in which agents only observe what their own actions let them see. When feedback is selective, learning can be permanently incomplete, and that is why stable play under incomplete information may reflect confirmed experience rather than full identification of the true model. (Wu et al., 2021) (Wu et al., 2022) (Fudenberg et al., 1993)

Evidence

(Wu et al., 2021)

Multi-agent Bayesian Learning with Best Response Dynamics: Convergence and Stability

> *"The notion of fixed point in our learning dynamics is similar to the self-confirming equilibrium introduced in (Fudenberg et al., 1993) for extensive-form games.1 At a self-confirming equilibrium, players maintain consistent beliefs of their opponents' strategies at information sets that are reached, but the beliefs of strategies can be incorrect at unreached information sets. Similarly, in our model, a fixed point can be different from a complete information equilibrium due to the incorrect estimates on the unobserved game outcomes formed by the beliefs (i.e., the estimated payoff distributions of strategies that differ from the fixed point may not be consistent). In general, these incorrect estimates may never be corrected by the learning dynamics because information of game outcomes is endogenously acquired based on the chosen strategies in each stage. 2"*

(Fudenberg et al., 1993)

Self-confirming equilibrium

> *"In a self-confining equilibrium, each player's strategy is a best response to his beliefs about the play of his opponents and each player's beliefs are correct along the equilibrium path of play. Thus, if a self-confirming equilibrium occurs repeatedly, no player ever observes play that contradicts his beliefs, even though beliefs about play at off-path information sets need not be correct. The authors characterize the ways in which self-confirming equilibria and Nash equilibria can differ and provide conditions under which self-confirming equilibria correspond to standard solution concepts. Copyright 1993 by The Econometric Society."*

(Wu et al., 2022)

Convergence and Stability of Coupled Belief-Strategy Learning Dynamics in Continuous Games

> *"Complete learning is not always guaranteed because fixed point beliefs may form incorrect estimates on the unobservable game outcomes, i.e. the payoffs of strategies that are not taken. These incorrect estimates may persist since the information of game outcomes used in belief updates is endogenously acquired based on strategies chosen by players"*
>
> *".The phenomenon that endogenous information acquisition leads to incomplete learning is central to a variety of settings that include"*
>
> *"self-confirming equilibrium learning in extensive form games (Fudenberg et al., 1993), (Fudenberg et al., 1995))."*